IMO Training – Algebraic Number Theory
Rings and Fields

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Introduction

Definition (Ideal)**.** A subset $I \subset R$ is an *ideal*, written $I \lhd R$, if

1. I is an additive subgroup of $(R, +, 0)$.
2. $a \in I, b \in R$ implies $a \cdot b \in I$.

Definition (Generator of ideal)**.** For an element $a \in R$, the *ideal generated by* a is

$$(a) = aR = \{a \cdot r : r \in R\} \lhd R$$

In general, let $a_1, \dots, a_k \in R$, the *ideal generated by* $a_1, \dots, a_k$ is

$$(a_1, \dots, a_k) = \{a_1 r_1 + \cdots + a_k r_k : r_1, \dots, r_k \in R\}$$

Definition (Principal ideal)**.** An ideal I is a *principal ideal* if $I = (a)$ for some $a \in R$.

Definition (Integral domain)**.** A non-zero ring R is an *integral domain* if for all $a, b \in R$, if $a \cdot b = 0$, then $a = 0$ or $b = 0$.

Definition (Zero divisor)**.** An element $x \in R$ is a *zero divisor* if $x \neq 0$ and there is a $y \neq 0$ such that $x \cdot y = 0 \in R$.

Definition (Field of fractions)**.** Let R be an integral domain. A *field of fractions* F of R is a field with the following properties:

1. $R \leq F$.
2. Every element of F may be written as $a \cdot b^{-1}$ for $a, b \in R$, where b^{-1} means the multiplicative inverse to $b \neq 0$ in F.

Remark. The set of rationals $\mathbb{Q}$ is the field of fractions of $\mathbb{Z}$.

Definition (Division)**.** For elements $a, b \in R$, we say a *divides* b, written $a \mid b$, if there exists $c \in R$ such that $b = ac$.

Definition (Associates)**.** $a, b \in R$ are *associates* if $a = bc$ for some unit c.

Definition (Irreducible)**.** $a \in R$ is *irreducible* if $a \neq 0$, a is not a unit, and if $a = xy$, then x or y is a unit.

Definition (Prime)**.** $a \in R$ is *prime* if a is non-zero, not a unit, and whenever $a \mid xy$, either $a \mid x$ or $a \mid y$.

Remark. If $r \in R$ is prime, then it is irreducible.

Definition (Euclidean domain)**.** An integral domain R is a *Euclidean domain* if there is a *Euclidean function* $\phi : R \setminus \{0\} \to \mathbb{Z}^+$ such that:

1. $\phi(a \cdot b) \geq \phi(b)$ for all $a, b \neq 0$
2. If $a, b \in R$, with $b \neq 0$, then there are $q, r \in R$ such that $a = b \cdot q + r$ and either $r = 0$ or $\phi(r) < \phi(b)$.

Definition (Principal ideal domain)**.** A ring R is a *principal ideal domain* (PID) if it is an integral domain, and every ideal is a principal ideal.

Definition (Unique factorization domain)**.** An integral domain R is a *unique factorization domain* (UFD) if

1. Every non-unit may be written as a product of irreducibles;
2. If $p_1 \cdot p_2 \cdot \cdots \cdot p_n = q_1 \cdot q_2 \cdot \cdots \cdot q_m$ with p_i, q_j irreducible, then $n = m$, and they can be reordered such that p_i is an associate of q_i.

Theorem. Let R be a principal ideal domain. Then R is a unique factorization domain.

Definition (Number Field)**.** A *number field* is a finite extension over $\mathbb{Q}$.

Definition (Algebraic Integer)**.** Let L be a number field. An *algebraic integer* is an $\alpha \in L$ such that there is some monic $f \in \mathbb{Z}[x]$ with $f(\alpha) = 0$. We write $\mathcal{O}_L$ for the set of algebraic integers in L.

Example. Find all ideals of $\mathbb{Z}$.

Proposition. Let R be a finite ring which is an integral domain. Then R is a field.

Proposition. R is a field iff its only ideals are $\{0\}$ and R.

Proposition. If $r \in R$ is prime, then it is irreducible.

Proposition. Let $I_1 \subset I_2 \subset \cdots$ be ideals in a ring R. Prove that $I = \bigcup I_i$ is also an ideal.

Example. Prove that the following is an Euclidean Domain (ED):

1. $\mathbb{Z}$
2. $\mathbb{Z}[i]$
3. $\mathbb{Z}[\omega]$

Proposition. Prove that if F is a field, then $F[X]$ is an ED.

Proposition. Prove that R is a field iff $R[X]$ is a PID.

Proposition. Prove that a field is an ED, and an ED is a PID.

Proposition. It is given that R is a PID. Prove that if $p \in R$ is irreducible, then it is prime.

Example. Prove that $\mathbb{Z}[\sqrt{d}]$ is not a UFD for $d = -3, -10, -13, -14$.

Example. Let $d > 1$ be an integer. Find all units in the ring $\mathbb{Z}[\sqrt{-d}]$. Also, find all units in the ring $\mathbb{Z}[\sqrt{2}]$.

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Proposition. Prove that a prime $p \in \mathbb{Z}$ is a prime in $\mathbb{Z}[i]$ iff $p \neq a^2 + b^2$ for $a, b \in \mathbb{Z}\backslash\{0\}$.

Proposition. It is given that $\mathbb{F}_p$ is a finite field. Prove that its multiplicative group is cyclic.

Proposition. Prove that all primes in $\mathbb{Z}[i]$ are either

(i) $p \equiv 3 \pmod 4$

(ii) $z \in \mathbb{Z}[i]$ with $N(z) = p$, where $p = 2$ or $p \equiv 1 \pmod 4$.

Example. Solve the following equations for $x, y \in \mathbb{Z}$:

1. $y^2 + 1 = x^3$
2. $y^2 + 2 = x^3$
3. $y^2 + 4 = x^3$
4. $y^2 + 11 = x^3$
5. $y^2 + y = x^3 - 2$

Proposition. Prove that $\mathcal{O}_{\mathbb{Q}} = \mathbb{Z}$.

Proposition. Let $\mathbb{Q} \subseteq K \subseteq L$ be finite extensions of fields. Prove that if $\alpha \in L$ is integral over $\mathcal{O}_K$, then it is an algebraic integer. Also, prove that if $f \in K[x]$ is monic, and $f^n \in \mathcal{O}_K[X]$ for some n, then $f \in \mathcal{O}_K[X]$.